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Propiedades de la dilatación isotrópica

Observación 1. Si $\phi \in \mathcal{L}^1(\mathbb{R}^n)$, entonces $\lim_{M\to\infty} \int_{\{\|x\|>M\}} \phi(x)\,\mathrm{d}x = 0$ ya que

$$\lim_{M\to\infty} \int_{\{\|x\|>M\}} \phi(x)\,\mathrm{d}x = \lim_{M\to\infty} \int_{\mathbb{R}^n} \phi(x)\mathbb{1}_{\{\|x\|>M\}}(x)\,\mathrm{d}x \overset{\substack{\text{TCD}\\ \downarrow}}{=} \int_{\mathbb{R}^n} \phi(x)\cdot 0\,\mathrm{d}x = 0.$$

Ahora sea $t>0$, definimos $\phi_t(x) := t^{-n}\phi\left(\frac{x}{t}\right)$. Entonces,

(1) $\displaystyle\int_{\mathbb{R}^n} \phi_t(x)\,\mathrm{d}x = \int_{\mathbb{R}^n} \frac{1}{t^n}\phi\left(\frac{x}{t}\right)\mathrm{d}x = \int_{\mathbb{R}^n} \phi(y)\,\mathrm{d}y$ mediante el cambio $y = \frac{x}{t}$.

(2) $\displaystyle\forall \eta > 0 : \lim_{t\to 0^+} \int_{\{\|x\|\geqslant\eta\}} \phi_t(x)\,\mathrm{d}x = \lim_{t\to 0^+} \int_{\{\|y\|\geqslant\eta/t\}} \phi(y)\,\mathrm{d}y = 0.$

Referenciado en

- `teo-aproximacion-identidad-convolucion`

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